ASSIGNMENT-5

1.Print the name in all possible patterns using function

[1]:
```
#Pattern
def row(name):
    for i in range(len(name)):
        for j in range(i+1):
            print(name[i],end=" ")
        print()
row('aakash')
```

```
a
a a
k k k
a a a a
s s s s s
h h h h h h
```

[14]:
```
def col(name):
    for i in range(len(name)):
        for j in range(5,i):
            print(" ",end=" ")
        for k in range(0,i+1):
            print(name[k],end=" ")
        print()
col("aakash")
```

```
a
a a
a a k
a a k a
a a k a s
a a k a s h
```

[2]:
```python
#pattern_lat_row
def row(name):
    for i in range(len(name)):
        for j in range(5,i,-1):
            print(" ",end=" ")
        for k in range(0,i+1):
            print(name[i],end=" ")
        print()
row("aakash")
```

```
          a
        a a
      k k k
    a a a a
  s s s s s
h h h h h h
```

[4]:
```python
#pattern_lat_col
def col(name):
    for i in range(len(name)):
        for j in range(5,i,-1):
            print(" ",end=" ")
        for k in range(0,i+1):
            print(name[k],end=" ")
        print()
col("aakash")
```

```
          a
        a a
      a a k
    a a k a
  a a k a s
a a k a s h
```

[7]:
```python
#pattern_pyname_row
def col(name):
    for i in range(len(name)):
        for j in range(5,i,-1):
            print("",end=" ")
        for k in range(0,i+1):
            print(name[i],end=" ")
        print()
col("aakash")
```

```
     a
    a a
   k k k
  a a a a
 s s s s s
h h h h h h
```

[8]:

```
#pattern_pyname_col
def col(name):
    for i in range(len(name)):
        for j in range(5,i,-1):
            print("",end=" ")
        for k in range(0,i+1):
            print(name[k],end=" ")
        print()
col("aakash")
```

```
     a
    a a
   a a k
  a a k a
 a a k a s
a a k a s h
```

[28]:

```
#Pattern_RAT_upperrow
def row():
    name="aakash".upper()
    for i in range(len(name)):
        for j in range(i+1):
            print(name[i],end=" ")
        print()
row()
```

```
A
A A
K K K
A A A A
S S S S S
H H H H H H
```

[33]:

```
#Pattern_RAT_uppercol
def col():
    name="aakash".upper()
    for i in range(len(name)):
        for j in range(i+1):
            print("",end="")
        for k in range (0,i+1):
            print(name[k],end=" ")
        print()
col()
```

```
A
A A
A A K
A A K A
A A K A S
A A K A S H
```

[34]:

```
#pattern_lat_row
def row():
    name='aakash'.upper()
    for i in range(len(name)):
        for j in range(5,i,-1):
            print(" ",end=" ")
        for k in range(0,i+1):
            print(name[i],end=" ")
        print()
row()
```

```
          A
        A A
      K K K
    A A A A
  S S S S S
H H H H H H
```

[35]:

```
#pattern_lat_col
def col():
    name='aakash'.upper()
    for i in range(len(name)):
        for j in range(5,i,-1):
            print(" ",end=" ")
        for k in range(0,i+1):
            print(name[k],end=" ")
        print()
col()
```

```
          A
        A A
      A A K
    A A K A
  A A K A S
A A K A S H
```

[37]:

```
#pattern_pyname_row
def row():
    name='aakash'.upper()
    for i in range(len(name)):
        for j in range(5,i,-1):
            print("",end=" ")
        for k in range(0,i+1):
            print(name[i],end=" ")
        print()
row()
```

```
     A
    A A
   K K K
  A A A A
 S S S S S
H H H H H H
```

[45]:

```
#pattern_pyname_row
def col():
    name='aakash'.upper()
    for i in range(len(name)):
        for j in range(5,i,-1):
            print("",end=" ")
        for k in range(0,i+1):
            print(name[k],end=" ")
        print()
col()
```

```
     A
    A A
   A A K
  A A K A
 A A K A S
A A K A S H
```

[48]:

```
#Patter_Ilat_col
def col():
    name='aakash'
    n=len(name)
    for i in range(n,0,-1):
        for j in range(n,i,-1):
            print(" ",end=" ")
        for k in range(0,i):
            print(name[k],end=" ")
        print()
col()
```

```
a a k a s h
  a a k a s
    a a k a
      a a k
        a a
          a
```

•[53]:
```
#Patter_Ilat_row
def row():
    name='aakash'
    n=len(name)
    for i in range(n,0,-1):
        for j in range(n,i,-1):
            print(" ",end=" ")
        for k in range(0,i):
            print(name[n-i],end=" ")
        print()
row()
```

```
a a a a a a
  a a a a a
    k k k k
      a a a
        s s
          h
```

[55]:
```
#Patter_Ilat_upperrow
def row():
    name='aakash'.upper()
    n=len(name)
    for i in range(n,0,-1):
        for j in range(n,i,-1):
            print(" ",end=" ")
        for k in range(0,i):
            print(name[n-i],end=" ")
        print()
row()
```

```
A A A A A A
  A A A A A
    K K K K
      A A A
        S S
          H
```

[56]:
```
#Patter_Ilat_uppercol
def col():
    name='aakash'.upper()
    n=len(name)
    for i in range(n,0,-1):
        for j in range(n,i,-1):
            print(" ",end=" ")
        for k in range(0,i):
            print(name[k],end=" ")
        print()
col()
```

```
A A K A S H
  A A K A S
    A A K A
      A A K
        A A
          A
```

2.Find the factors of given 2 digit number using function

[57]:
```
def num_fac():
    n=int(input("Enter the number:"))
    for i in range(1,n+1):
        if n%i==0:
            print(i)
num_fac()
```

```
Enter the number: 20
1
2
4
5
10
20
```

[58]:
```
def num_fac():
    n=int(input("Enter the number:"))
    for i in range(1,n+1):
        if n%i==0:
            print(i)
num_fac()
```

```
Enter the number: 99
1
3
9
11
33
99
```

3.Find the biggest number using lambda

```
[67]: num=lambda a,b,c,d:a if a>b and a>c and a>d else b if b>c and b>d else c if c>d else d
num(10,20,30,40)
```

```
[67]: 40
```

```
[68]: def max(a,b,c,d):
    num=lambda a,b,c,d:a if a>b and a>c and a>d else b if b>c and b>d else c if c>d else d
    print(num(a,b,c,d))
max(100,250,50,150)
```

```
250
```

```
[69]: def max(a=500,b=200,c=700,d=100):
    num=lambda a,b,c,d:a if a>b and a>c and a>d else b if b>c and b>d else c if c>d else d
    print(num(a,b,c,d))
max()
```

```
700
```

4.Create a list for cube from the list and store the cube value using lambda using function

```
[70]:
num=[5,7,8,2,4]
cube=list(map(lambda a:a**3,num))
print(cube)
```

```
[125, 343, 512, 8, 64]
```

```
[72]: def cube(num):
    q=list(map(lambda a:a**3,num))
    print(q)
num=[10,20,30,40]
print(cube(num))
```

```
[1000, 8000, 27000, 64000]
None
```

```
[75]: def cube(num=[10,20,30,40]):
    q=list(map(lambda a:a**3,num))
    print(q)

cube()
```

```
[1000, 8000, 27000, 64000]
```